

Case Study no. 39

Coppervale Semiconductor Fab

Network Outage Investigation — Capstone Diagnostics

Problem Statement:

Coppervale Semiconductor Fab runs three sites — a Head Office (HQ), a Wafer Fabrication Plant still running a legacy EIGRP network, and a Test & Packaging Facility — tied together by an OSPF backbone and dual-ISP BGP Internet access. Over the past 48 hours, staff at every site reported intermittent problems. The network team captured a set of symptoms but has not yet found root causes. You are the on-call engineering team: diagnose each symptom, identify the most likely root cause, and propose the fix and a verification step for each.

Sym.	Location	Observed Behaviour
A	Fab Plant access layer	A switch port configured with Root Guard enters a root-inconsistent (blocking) state; logs show the port received a superior BPDU from a neighboring switch whose priority was recently, and apparently mistakenly, lowered below the intended root bridge's.
B	HQ–Fab Plant OSPF link (multipoint)	On a multipoint WAN segment, one router's OSPF neighbor entry for a specific remote router is stuck in the 'ATTEMPT' state and never progresses, while its other neighbors on the same segment are fully adjacent.

Sym. Location Observed Behaviour

A Fab Plant access layer A switch port configured with Root Guard enters a root-inconsistent (blocking) state; logs show the port received a superior BPDU from a neighboring switch whose priority was recently, and apparently mistakenly, lowered below the intended root bridge's.

B HQ–Fab Plant OSPF link (multipoint) On a multipoint WAN segment, one router's OSPF neighbor entry for a specific remote router is stuck in the 'ATTEMPT' state and never progresses, while its other neighbors on the same segment are fully adjacent.

C Fab Plant–Test & Packaging EIGRP route A router recently configured as an EIGRP stub is not advertising a route that the rest of the network expects it to provide, even though the route exists in its local routing table.

D HQ Internet edge (BGP) The eBGP session to the secondary ISP repeatedly resets; logs show the session was torn down immediately after the router logged a message about exceeding a configured maximum-prefix limit.

E Fab Plant access layer (Wireshark) A packet capture shows a burst of unusually large, fragmented ICMP echo request packets arriving in rapid succession, all targeting a single internal host.

Sym.	Location	Observed Behaviour
C	Fab Plant–Test & Packaging EIGRP route	A router recently configured as an EIGRP stub is not advertising a route that the rest of the network expects it to provide, even though the route exists in its local routing table.
D	HQ Internet edge (BGP)	The eBGP session to the secondary ISP repeatedly resets; logs show the session was torn down immediately after the router logged a message about exceeding a configured maximum-prefix limit.
E	Fab Plant access layer (Wireshark)	A packet capture shows a burst of unusually large, fragmented ICMP echo request packets arriving in rapid succession, all targeting a single internal host.

Objectives

1. For Symptom A, explain what Root Guard is designed to prevent, why a neighboring switch sending a superior BPDU triggers the root-inconsistent state, and state the specific commands/output you would use to confirm the cause and restore the port once the misconfigured priority is corrected.
2. For Symptom B, explain why an OSPF neighbor on a multipoint (NBMA-style) segment can remain stuck in ATTEMPT state (consider a missing static neighbor statement for that specific router) while other neighbors on the same segment adjacency normally, and state the fix.
3. For Symptom C, explain the difference between EIGRP stub types (e.g., a stub configured as receive-only versus one permitted to advertise connected/summary routes), why an incorrect stub type would silently withhold an expected route, and state the fix.
4. For Symptom D, explain what the BGP maximum-prefix limit is for, why exceeding it causes the session to be torn down rather than simply ignoring extra routes, and the commands you would use to check the configured limit, understand why more routes than expected were received, and safely re-establish the session.
5. For Symptom E, state what the capture pattern indicates, its likely intent, and two concrete mitigations (e.g., rate-limiting ICMP, ACLs) that would help prevent or limit this kind of activity.
6. Considering all five symptoms together, decide whether they represent one common trigger (e.g., a single recent change at the Fab Plant) or more than one unrelated fault. Justify your conclusion using evidence from the symptom log.

7. Produce a single root-cause report ranking the five symptoms by business impact and proposing an order of remediation, with a verification command or check for each fix.

Outcomes

- Diagnose the network symptoms systematically using appropriate networking and security concepts.
- Identify likely root causes, propose technically appropriate corrective actions, and prioritize remediation.
- Define verification steps that demonstrate whether each corrective action has resolved the observed problem.

Deliverables

- A root-cause analysis table: Symptom → Likely Cause → Evidence → Fix → Verification Step.
- A short written narrative (300–500 words) addressing whether the symptoms are related.
- A prioritized remediation plan with business-impact justification for the ordering.
- Annotated ‘before/after’ command output (real or representative) showing each fix verified.